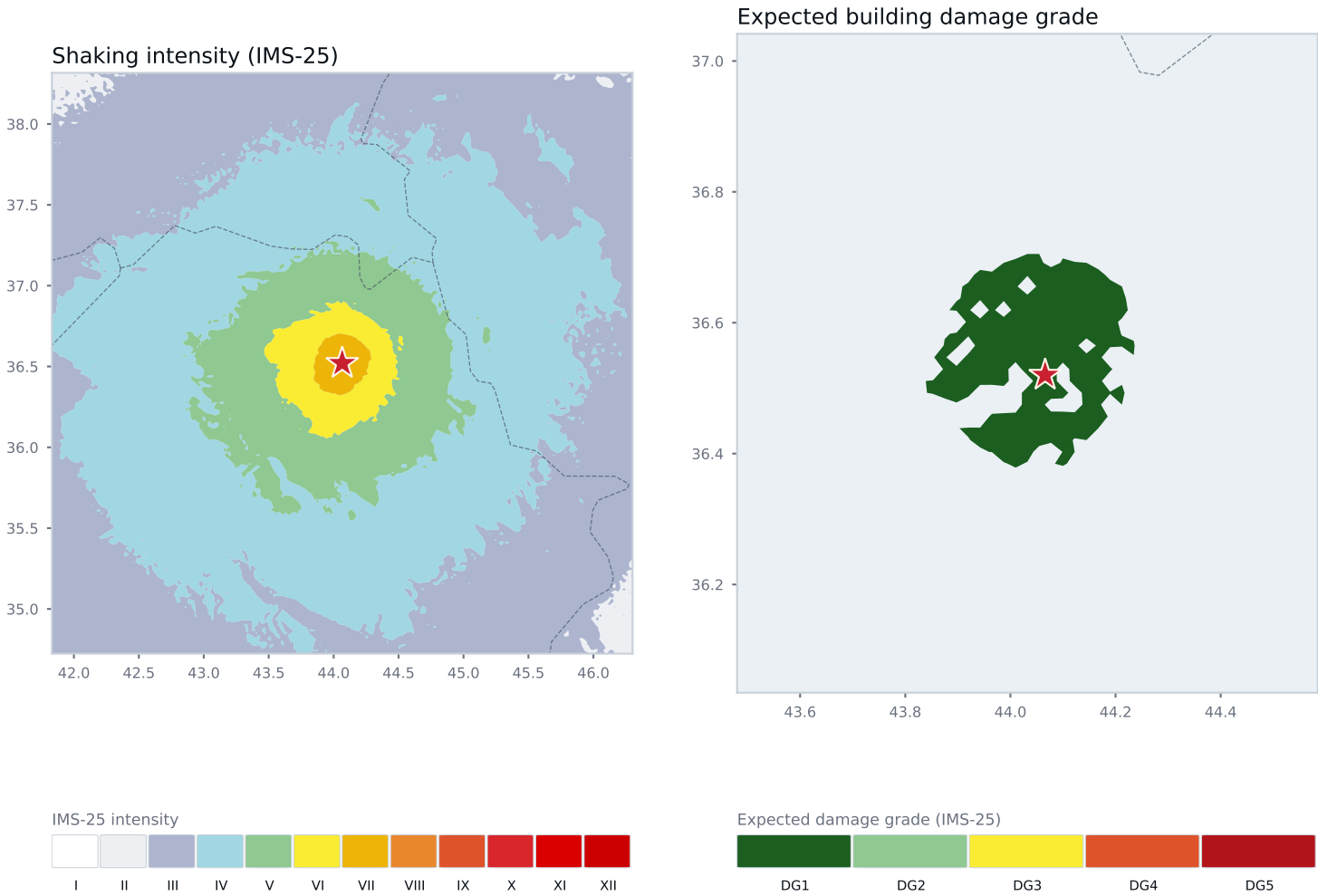




Damage alert

ORANGE

Widespread damage; many buildings unusable

Provisional

Day occupancy

Buildings heavily damaged (DG3 or worse)

1.8 thousand to 39 thousand 3.4 million

Central estimate 9.5 thousand. The range is the 5th to 95th percentile.

Buildings in the shaken area

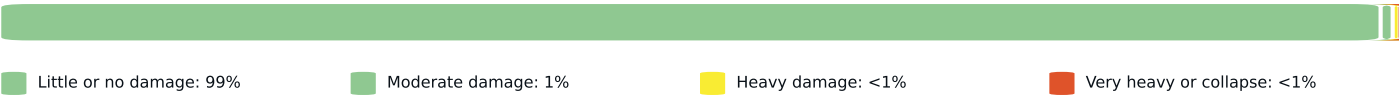
21% of them inferred from built-up area, not observed footprints.

People by shaking level



Building exposure modelled for: Iraq. Shaded areas without exposure data are shown for shaking only.

Share of buildings in the mapped area by expected damage

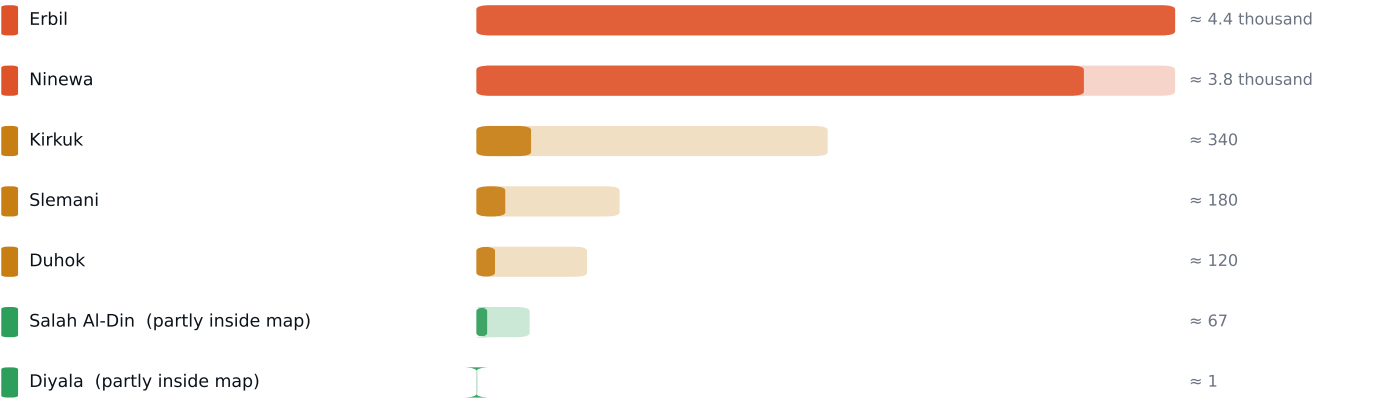


Which buildings failed

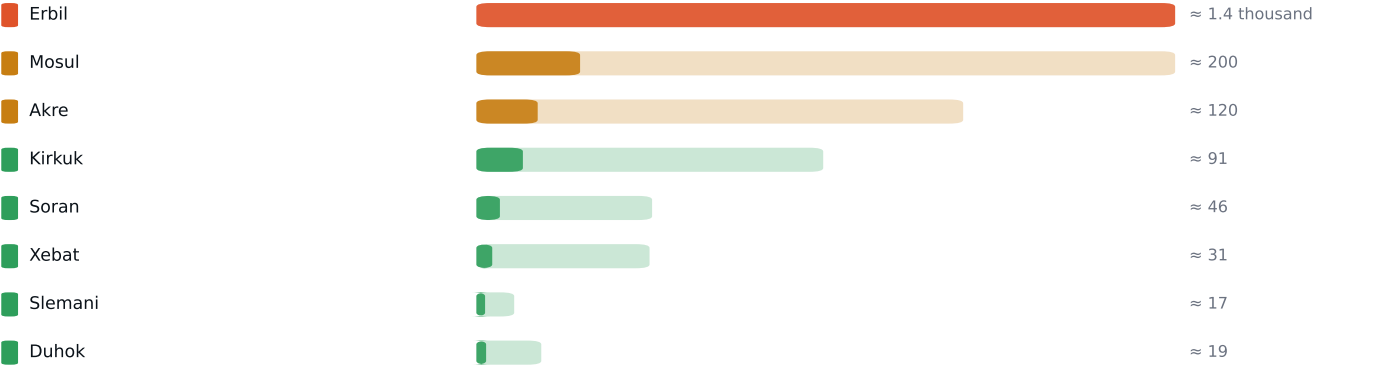
Share of each type's own buildings heavily damaged, worst first



Governorates most affected: heavily damaged buildings, central estimate and 90% range



Cities most affected



Every governorate inside the map

Area	Buildings	Heavy (P50)	90% range	% of its stock
Erbil	659 thousand	2.4 thousand	740 to 22 thousand	0.36%
Ninewa	960 thousand	2.1 thousand	710 to 16 thousand	0.21%
Kirkuk	570 thousand	190	69 to 2.2 thousand	0.03%
Slemani	632 thousand	110	43 to 890	0.02%
Duhok	266 thousand	69	27 to 690	0.03%
Salah Al-Din	261 thousand	45	18 to 330	0.02%
Diyala	11 thousand	1	none to 8	0.01%

How this report was made

Hazard engine	Bumelerze SHAKEmap 26.1.1, ground-motion models CY14, ASB14, BSSA14, KALE15, intensity via Zaniniandhofer19 (IMS-25)
Conditioning	none (catalog only)
Observations	0 seismic stations, 0 felt-report cells (felt channel: none)
Distances	ps2ff; rupture: Origin
Damage scale	IMS-25
Provenance	config 09a6a04fa72f, product version 1, event bml19910001

Data sources

Earthquake parameters	USGS, EMSC, GEOFON
Building footprints	GFZ OpenBuildingMap, from OpenStreetMap (c) OpenStreetMap contributors, ODbL; Microsoft Building Footprints, CDLA-Permissive-2.0
Built-up surface, settlements	European Commission JRC, Global Human Settlement Layer, CC BY 4.0
Population	WorldPop, University of Southampton, CC BY 4.0
Administrative boundaries	OCHA COD-AB Iraq, CC BY-IGO; source authority Iraq Central Statistical Organisation
Building types, vulnerability	GEM Global Exposure Model; IMS-25 vulnerability table
Casualty rates (computed, not published)	FEMA Hazus Earthquake Model Technical Manual 5.1

What this is, and what it is not

An automatic estimate of shaking and building damage from the earthquake's location and magnitude, adjusted with whatever station and felt-report data existed at the time.

Cite as: Bumelerze SHAKEmap and damage report, event bml19910001, version 1, generated 21 Sep 2026, 22:18 UTC. CC BY 4.0.